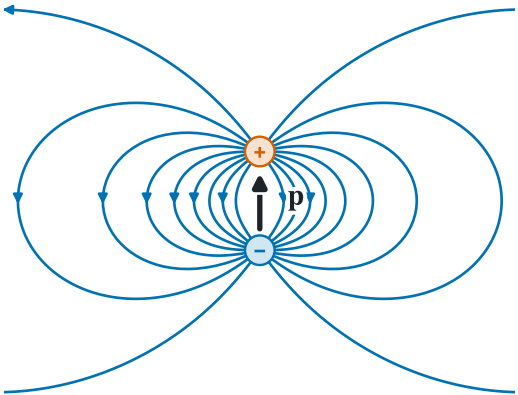
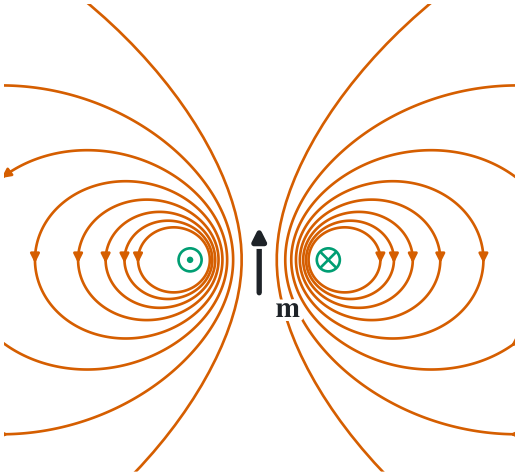
Electric dipole: a ± charge pair



between the charges: E points down, against p

Magnetic dipole: a small current loop



inside the loop: B points up, along m

Far away the two patterns coincide (both are dipole fields); near the source, E lines start and end on charges, B lines close on themselves.